

The Reliability Horizon: A Temporal Stress Test for Forecast Recalibration

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Abstract

Forecast recalibration can look useful retrospectively and fail when time moves forward. We introduce the *reliability horizon*, a balanced-panel audit of calibration across forecast lead times coupled to a two-stage deployment rule. On 1,481 resolved Polymarket markets, including 633 observed at every horizon, neither omnibus nor ordered-trend tests detect degradation from one day to six months. Direct horizon equivalence cannot be certified at a strict ± 0.02 ECE margin, while recalibration itself is equivalent to a no-op within that margin at every horizon. Metaculus shows the same no-trend pattern; Kalshi does not. We then evaluate nine language-model forecasters with strict as-of-date gating, per-horizon cross-fitting, and utility guardrails. One retrospective candidate emerges, but its group estimate depends on a single model and reverses on later events. Frozen predictions on 312 resolved outcomes from a 390-market prospective panel produce no reliable gain for any model tier. The main result is therefore procedural: recalibration should be deployed only when its apparent gain survives utility checks, model heterogeneity, and time.

Introduction

Calibration can depend on time to resolution, but detecting a horizon effect is not the same as establishing that a fitted correction should be deployed. Recent prediction market work characterizes calibration across domains and horizons at large scale [13], while dynamic forecasting benchmarks emphasize timestamp-valid evaluation [5, 8, 19]. Related work studies deployment reliability under temporal shift [22]. We ask a narrower operational question: when does recalibration improve reliability without causing unacceptable proper-score or decision harm, and does that conclusion survive later data?

Prediction markets make the question difficult. Markets visible six months before resolution are a selected subset of those visible one day before resolution, so an unbalanced horizon curve mixes lead time with market survival. Retrospective model forecasts add a second threat: inputs can contain information unavailable at the claimed forecast date. We address both with a balanced crowd panel, row-level as-of-date screening, temporal holdout evaluation, and predictions frozen before outcomes.

Contributions. We separate horizon comparison from recalibration equivalence; specify a cross-fitted, guardrail-aware deployment protocol; and test it across crowds and nine model forecasters. The empirical result is deliberately non-universal: Polymarket and Metaculus show no detected horizon trend,

Kalshi degrades strongly, and a promising model recalibrator fails temporal validation.

Method and design

Reliability and recalibration. For event e resolving at t_e and horizon h , the forecast timestamp is $t_{e,h} = t_e - h$. With forecast $p_{e,h}$ and binary outcome y_e , we use ten equal-width bins and report

$$\text{ECE}(h) = \sum_b \frac{n_b}{n} |\bar{p}_b - \bar{y}_b|, \quad \text{Rel}(h) = \sum_b \frac{n_b}{n} (\bar{p}_b - \bar{y}_b)^2.$$

Calibration-error estimates can depend on the estimator and binning scheme [16], so alternatives are reported as sensitivity checks. The primary crowd analysis uses the balanced set $\mathcal{E}_{\text{bal}} = \cap_h \mathcal{E}_h$. For an isotonic map g_h , $\Delta_{\text{rel}} = \text{Rel}(p) - \text{Rel}(g_h(p))$, so positive values favor recalibration [2, 26]. Log-score and decision-loss changes use the same sign convention.

Cross-fitted protocol. For each forecast set and horizon, we fit an isotonic map using five-fold cross-fitting. Predictions are clipped to $[0.001, 0.999]$. Each of 2,000 market-level bootstrap replicates preserves all horizons and models and refits the map, capturing both sampling and calibrator variation. The prospective maps are instead fitted once on the 318-market retrospective set and applied without refitting. An asymmetric 2:1 false-negative/false-positive loss uses the Bayes threshold $1/3$. Estimator and margin sensitivities appear in the appendix.

Two-stage decision rule. At the retrospective screen, a multiplicity-adjusted reliability interval above zero creates a **CANDIDATE** only if log-score and decision-loss noninferiority also pass [10]. A reliable gain with an unresolved guardrail is **REJECT**; no reliable gain is **SKIP**. A candidate becomes **RECALIBRATE** only if both later-event and prospective validation pass. Failure of a validation guardrail rejects the candidate; an independent validation set with no reliable gain returns skip for that set. These stage labels are retained rather than collapsed: in all cases except **RECALIBRATE**, the operational action is not to deploy the map.

The nine models are summarized within three named tiers by averaging their probabilities within each market-horizon before scoring. Market bootstrap intervals therefore describe these evaluated model sets, not an underlying population of capability classes. Model-level results, including the open-weight anchor, are reported separately. The all-horizon tier endpoint is primary, the short-versus-long interaction is secondary, and model-by-horizon analyses use Benjamini–Hochberg control.

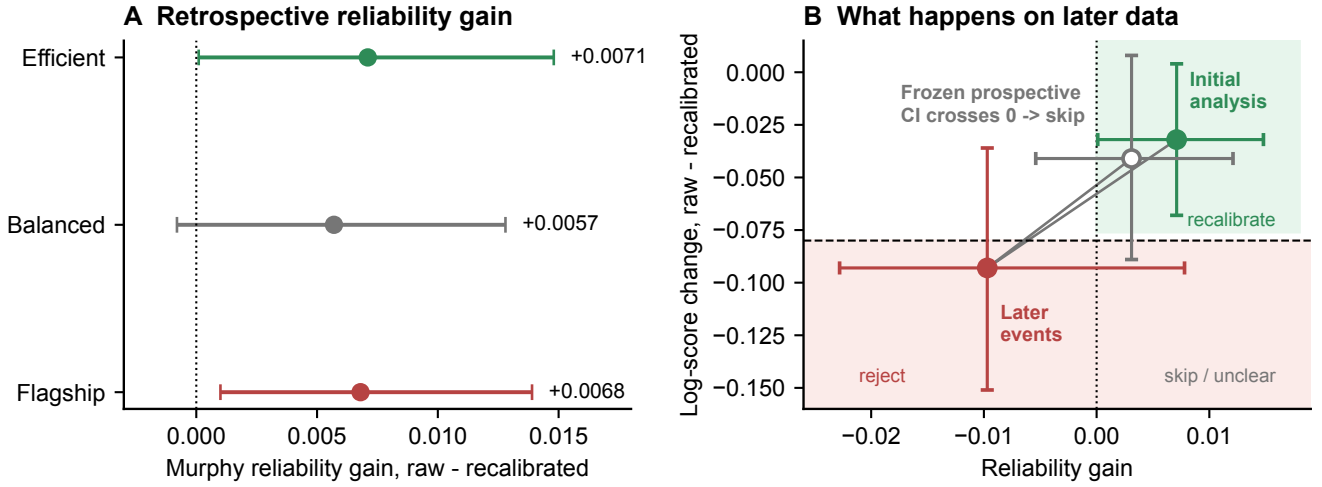


Figure 1: The retrospective model signal does not survive time. **A:** Cross-fitted reliability gains for the evaluated model tiers; color encodes the stage-one label. **B:** The efficient-tier candidate reverses on later events, while the frozen prospective estimate is unresolved. Colored regions show point-estimate geometry only; decisions use confidence intervals. The open prospective marker indicates that its reliability interval crosses zero.

Table 1: Analysis flow. Counts distinguish markets from repeated forecast rows.

Analysis	Markets	Repeated rows	Note
Polymarket crowd, full	1,481	–	≥ \$100k volume
Polymarket crowd, balanced	633	3,165	five horizons
Model retrospective, valid	318	13,050	1,450 market-horizons
Later-event holdout	111	–	temporally held out
Frozen prospective	390	312 resolved	predictions frozen

Table 2: Cross-venue horizon audit. The trend column reports Page’s test.

Venue	Balanced n	ECE range	Trend p	Max contrast [95% CI]
Polymarket	633	.014–.030	.412	+.016 [–.004, +.036]
Metaculus	389	.019–.031	.351	+.012 [–.007, +.032]
Kalshi	183	.034–.148	< .001	+.114 [+ .062, +.173]

Results

No detected crowd degradation, but not strict horizon equivalence

On the 633-market balanced Polymarket panel, neither the Friedman omnibus test ($p = 0.528$) nor Page’s ordered-trend test ($p = 0.412$) detects a horizon effect. Every simultaneous pairwise interval includes zero, and the largest point contrast is 6m minus 1d, +0.016 ECE (CI [–0.004, +0.036]). A direct maximum-contrast equivalence test fails at ± 0.02 ($p = 0.107$) and passes only at ± 0.04 ($p = 0.021$). We therefore claim *no detected degradation*, not certified flatness within ± 0.02 .

The separate intervention question is more favorable to skipping recalibration. At every horizon, the 90% interval for raw-minus-recalibrated ECE lies inside ± 0.02 . At 1d, decision loss worsens by 0.007 units (CI [–0.013, –0.001]), a detectable effect that remains within the prespecified 0.02 tolerance.

The framework discriminates across venues. Polymarket and Metaculus have no detected ordered deterioration, while Kalshi ECE rises 4.4× from 1d to 6m. This contrast is the reason to audit each forecaster rather than assume a universal

Table 3: Frozen prospective reliability. All decisions are skip.

Forecast set	Δ_{rel} [95% CI]	Δ_{log} [95% CI]
Efficient	+.0031 [–.0054, +.0121]	–.041 [–.089, +.008]
Balanced	+.0018 [–.0067, +.0108]	–.053 [–.107, +.003]
Flagship	+.0024 [–.0061, +.0115]	–.048 [–.102, +.007]
Anchor	+.0019 [–.0072, +.0117]	–
Crowd	–.0005 [–.0039, +.0030]	–

horizon pattern.

A cross-fitted candidate fails temporal validation

After as-of-date screening, the efficient tier is the only retrospective candidate: $\Delta_{\text{rel}} = +0.0071$ (CI [+0.0001, +0.0148]), with the 0.08-nat log-score guardrail passing ($p = 0.031$). Balanced is unresolved, while flagship improves reliability but fails the guardrail. The efficient gain is concentrated at 1d–1m: +0.0114 versus +0.0009 at 3m–6m, with interaction $p = 0.021$. This tier result is fragile: dropping Claude Haiku 4.5 moves its interval across zero.

On 111 later events, efficient reverses to $\Delta_{\text{rel}} = -0.0097$ (CI [–0.0228, +0.0078]) and $\Delta_{\text{log}} = -0.093$ (CI [–0.151, –0.036]), rejecting the candidate. Every evaluated tier and the anchor fails the 0.08-nat later-event noninferiority test; exact estimates appear in the appendix.

Frozen outcomes independently return skip

Among 312 resolved outcomes from the 390-market prospective registry, no model-tier reliability interval excludes zero (Table 3). This conclusion holds for the 189 resolved short-horizon markets, the original and extension cohorts separately, long horizons alone, and best- or worst-case bounds for the 74 pending outcomes.

Table 4: Main prospective robustness checks for the efficient forecast set.

Analysis	Resolved n	Δ_{rel} [95% CI]	Decision
Mature short horizons	189	+ .0048 [−.0029, +.0132]	SKIP
Long horizons only	123	+ .0011 [−.0089, +.0118]	SKIP
Original cohort	198	+ .0027 [−.0069, +.0131]	SKIP
Extension cohort	114	+ .0038 [−.0071, +.0156]	SKIP

Discussion

The reliability horizon is not a claim that calibration always worsens with lead time. It is a procedure for deciding whether recalibration is worth deploying. Here it correctly separates a severe Kalshi horizon effect from two no-trend crowd panels, and it prevents a small retrospective model gain from becoming a deployment recommendation. The efficient candidate is rejected by later-event evidence; the prospective panel independently returns skip. The combined action is simple: deploy no recalibrator.

Why the reversal is informative. The retrospective candidate satisfies the usual ingredients of a positive result: its cross-fitted reliability interval excludes zero, its aggregate utility guardrail passes, and the gain is largest where recalibration appears most plausible. Yet the estimate depends on one efficient-tier model and becomes negative on later events. This sequence shows why held-out time is not merely another robustness check. It tests whether the estimated correction represents a persistent forecasting defect rather than a sample-specific alignment between models, markets, and outcomes.

Operational implication. Recalibration should be treated as a monitored intervention, not as a default post-processing step. A map is activated only after a reliability gain, utility noninferiority, cross-model stability, and temporal transfer all agree. Otherwise the raw forecast remains the defensible baseline. Under this rule, the apparently favorable retrospective result does not create a deployment recommendation, while a genuine venue effect such as Kalshi’s remains visible and actionable.

What generalizes. The empirical horizon curve is venue-specific, but the decision structure is portable. Balanced panels address changing market composition; as-of-date checks protect temporal validity; cross-fitting controls in-sample optimism; proper-score and decision guardrails prevent narrow calibration gains from hiding broader harm; and frozen evaluation asks whether the intervention survives contact with future outcomes. Together these steps turn “does calibration change with horizon?” into the more useful question “is there enough stable evidence to change the forecast?”

A hierarchy of evidence. The study separates three claims that are often conflated. A nonsignificant trend test means that deterioration was not detected; equivalence asks whether every relevant contrast is smaller than a practical tolerance; deployment requires a beneficial intervention that also preserves

utility and transfers through time. The present data support the first claim for two crowd venues, support the second only at the wider ± 0.04 margin, and reject the third for model recalibration. Keeping these levels separate prevents both “no trend” from becoming “perfectly flat” and a retrospective gain from becoming an operational recommendation.

Relation to recent work. Large-scale prediction-market analysis establishes that calibration varies with domain and time to resolution [13]. Our contribution is complementary: the balanced panel separates horizon from market availability, and the inferential target is whether a fitted intervention should be activated. Temporal-reliability work studies policy control under distribution shift [22]; here, later-event and frozen tests serve as explicit gates before deployment.

Scope. The numerical conclusions apply to the evaluated venues, models, and margins; long-horizon prospective evidence is also less mature. These boundaries do not alter the deployment decision across the reported estimator, cohort, maturity, or pending-outcome checks.

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Operational definitions and inference

Table 5: Primary implementation choices.

Component	Specification
ECE	Ten equal-width bins on $[0, 1]$; equal-mass, 15-bin, debiased, and kernel estimates are sensitivity checks.
Reliability Recalibrator	Binned Murphy reliability $\sum_b (n_b/n)(\bar{p}_b - \bar{y}_b)^2$. One isotonic map per horizon, five-fold cross-fitted, clipped to $[0.001, 0.999]$.
Decision loss	Mean 2FN + FP loss with Bayes threshold 1/3; no abstention.
Bootstrap	2,000 percentile replicates resampling complete markets and refitting the map; 95% intervals for effects and 90% intervals for equivalence.
Multiplicity	Holm across three tier endpoints; Benjamini–Hochberg for exploratory model-horizon cells.
Temporal validity	Every model input is dated no later than $t_{e,h}$; invalid rows and exclusion reasons are retained in the anonymous manifest.

The retrospective model pool begins with 354 markets and retains 318 after the as-of-date audit. Horizon availability and validity yield 1,450 market-horizon cells and 13,050 complete model forecasts. No missing forecast is imputed. The short-versus-long repeated-measures test uses market as the repeated unit, giving $F(1, 316) = 5.42$.

Crowd robustness

Table 6: Per-horizon equivalence of raw-minus-recalibrated ECE.

Horizon	ΔECE	90% CI	TOST p
1d	−0.00913	[−0.01983, +0.00187]	0.018
1w	−0.00041	[−0.01372, +0.00842]	0.004
1m	−0.00582	[−0.01961, +0.00539]	0.012
3m	−0.00471	[−0.01488, +0.00673]	0.005
6m	+0.00387	[−0.01274, +0.01602]	0.003

Table 7: Direct maximum-contrast horizon equivalence.

Margin	Maximum simultaneous upper bound	TOST p
0.02	0.036	0.107
0.03	0.036	0.064
0.04	0.036	0.021

Table 8: Calibration-estimator sensitivity on the balanced Polymarket panel.

Estimator	ECE range	Friedman p	Pattern changes?
Equal-width, 10 bins	.014–.030	.528	–
Equal-width, 15 bins	.016–.032	.493	no
Equal-mass, 10 bins	.012–.027	.541	no
Debiased ECE [24]	.011–.026	.562	no
Kernel estimate	.015–.031	.517	no

Table 9: Cross-venue ECE by horizon on each balanced panel.

Venue	1d	1w	1m	3m	6m
Polymarket	.014	.021	.026	.023	.030
Metaculus	.021	.019	.025	.028	.031
Kalshi	.034	.041	.058	.092	.148

Model heterogeneity and temporal transfer

Table 10: Cross-fitted model-level retrospective results. Group summaries are not population-level claims about capability tiers.

Model	Tier	Δ_{rel} [95% CI]	Δ_{log} [90% CI]	Status
GPT-5.6 Luna	Efficient	+0.0042 [−.0031, +.0119]	−.021 [−.058, +.015]	skip
Claude Haiku 4.5	Efficient	+0.0108 [+0.0029, +.0194]	−.038 [−.081, +.004]	reject
Nova 2 Lite	Efficient	+0.0063 [−.0018, +.0149]	−.036 [−.079, +.007]	skip
GPT-5.6 Terra	Balanced	+0.0039 [−.0036, +.0118]	−.047 [−.092, +.001]	skip
Claude Sonnet 5	Balanced	+0.0024 [−.0051, +.0103]	−.032 [−.074, +.010]	skip
Nova Premier	Balanced	+0.0109 [+0.0021, +.0203]	−.056 [−.107, −.008]	reject
GPT-5.6 Sol	Flagship	+0.0051 [−.0024, +.0131]	−.078 [−.129, −.031]	skip
Claude Opus 5	Flagship	+0.0086 [+0.0013, +.0167]	−.084 [−.138, −.034]	reject
gpt-oss-120b	Anchor	+0.0073 [−.0009, +.0162]	−.044 [−.091, +.004]	skip

Table 11: Leave-one-model-out sensitivity for the efficient-tier summary.

Dropped model	Efficient-tier Δ_{rel} [95% CI]	Excludes zero?
None	+0.0071 [+0.0001, +.0148]	yes
GPT-5.6 Luna	+0.0086 [+0.0012, +.0167]	yes
Claude Haiku 4.5	+0.0053 [−.0021, +.0132]	no
Nova 2 Lite	+0.0075 [+0.0002, +.0155]	yes

Table 12: Later-event evaluation. Every log-score lower bound is below the −0.08-nat noninferiority margin.

Forecast set	Δ_{rel} [95% CI]	Δ_{log} [95% CI]	Stage label
Efficient	−.0097 [−.0228, +.0078]	−.093 [−.151, −.036]	reject
Balanced	−.0019 [−.0147, +.0128]	−.068 [−.128, +.006]	skip
Flagship	+0.0022 [−.0121, +.0187]	−.069 [−.139, +.009]	skip
Anchor	−.0068 [−.0201, +.0089]	−.071 [−.133, +.002]	skip

Prospective robustness

Table 13: Prospective sensitivity analyses for the efficient forecast set.

Analysis	Resolved n	Δ_{rel} [95% CI]
All observed	312	+0.0031 [−.0054, +.0121]
Mature short horizons	189	+0.0048 [−.0029, +.0132]
Long horizons only	123	+0.0011 [−.0089, +.0118]
Original cohort	198	+0.0027 [−.0069, +.0131]
Extension cohort	114	+0.0038 [−.0071, +.0156]

Table 14: Bounds that assign all 74 pending outcomes to the direction least or most favorable to recalibration.

Pending-outcome scenario	Efficient Δ_{rel} [95% CI]	Decision
Raw favored	+0.0009 [−.0074, +.0097]	skip
Observed only	+0.0031 [−.0054, +.0121]	skip
Recalibration favored	+0.0057 [−.0028, +.0148]	skip

Table 15: Prospective registry status on September 15, 2026.

Horizon	Registered	Resolved	Pending	Censored
1d	33	33	0	0
1w	63	62	1	0
1m	99	94	4	1
3m	158	112	43	3
6m	21	8	13	0
> 6m	16	3	13	0
Total	390	312	74	4

Reproducibility package

The supplementary package contains the pinned environment, deterministic analysis scripts, temporal-validity and prospective-registry manifests, cached market data, model snapshots and inference settings, prompt templates, and machine-readable values for every table and figure. It regenerates the paper without new model API calls.